Estimating latent factors: the Kalman filter

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Abstract

In a 1988 working paper, Stock and Watson proposed a way to estimate and forecast the GDP for the US in the short run. The model adopted in their paper is based on the notion that the comovements in many macroeconomic variables that can be captured by a single, underlying, unobserved variable. Therefore, they assumed that the four indicators they use form a dynamic one-factor model with the latent factor that represents economic activity and that, therefore, can be mapped to GDP.

The motive behind this project is to test the model proposed by S&W, verify that the assumptions of one-factor models hold, and test its performance under non-Gaussian distributions, which are common with macroeconomic data.

Then, we will try to feed the model more variables, and apply Principal Component Analysis (PCA) to reduce the number of variables such that we still preserve only one latent factor.

1. The Stock & Watson model

First, we test the model proposed by Stock and Watson. The vector of indicators we will be using is the following:

$$y_t = \begin{bmatrix} y_{1t} \\ y_{2t} \\ y_{3t} \\ y_{4t} \end{bmatrix}$$

where each one of the macroeconomic indicators is represented in log-differences to capture growth rates, and is standardized to adjust for the scale.

The factor we want to extract is the scalar f_t, which follows an AR(2) process. The complete model is therefore given by:

- The measurement equation: $y_t = \lambda f_t + e_t$
- The transition equation: $f_t = \phi f_{t-1} + w_t$

with:

$$f_t = \phi_1 f_{t-1} + \phi_2 f_{t-2} + w_t, \quad w_t \sim i.i.d. N(0,1)$$

$$e_{it} = \psi_{i1} e_{i,t-1} + \psi_{i2} e_{i,t-2} + \epsilon_{it}, \quad \epsilon_{it} \sim i.i.d. N(0, \sigma_i^2)$$

add graphical model

The Kalman filter requires errors e_t to be white noise normal. Given that the errors are not white noise, but follow an AR(2) process, they are incorporated into the state vector of the Kalman filter so that the filter's transition matrix is able to track the autocorrelated memory of the error.

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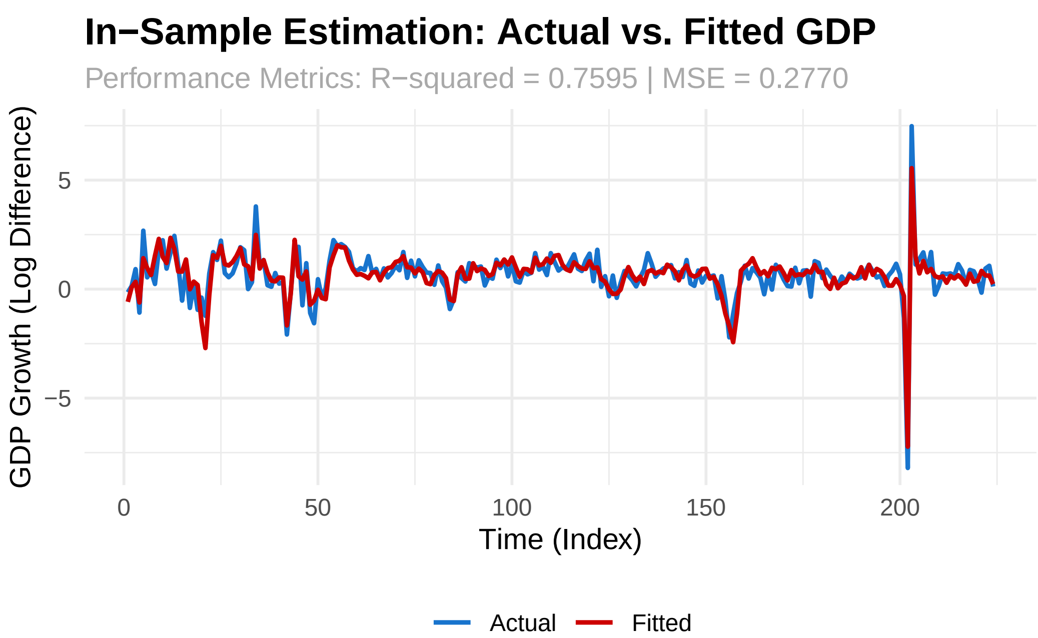
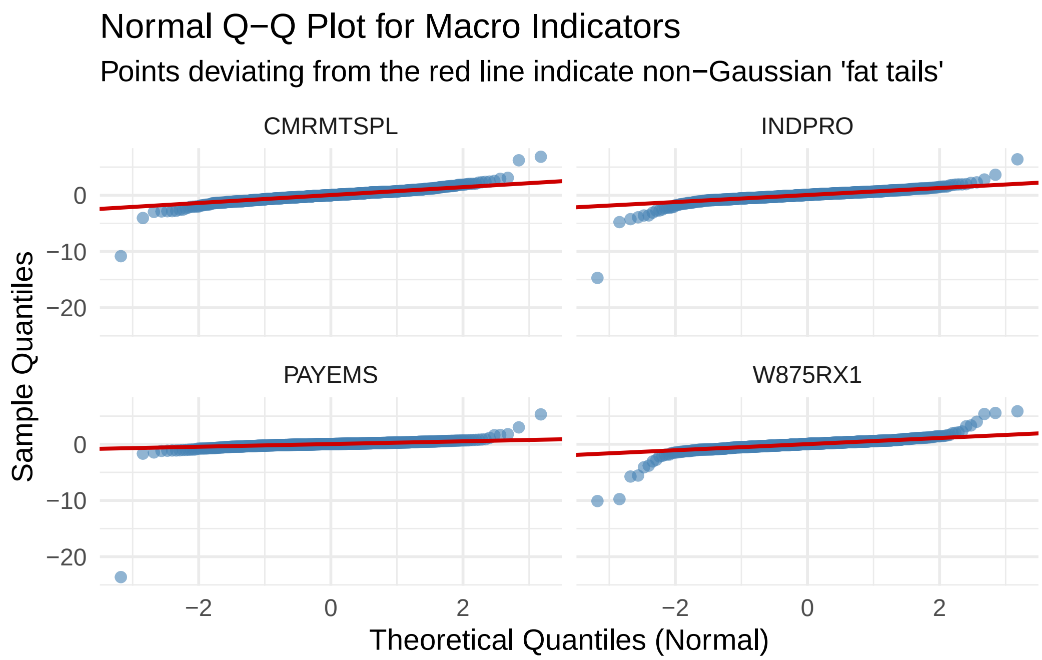
1.2. Extracting the factor

To extract the factor we run the Kalman filter.



1.3. The normality assumption and the tetrad costraints

The dynamic factor model proposed by Stock and Watson assumes gaussianity. However, given the fairly common shocks of the economy, it is hard to believe that macroeconomic indicators are normally distributed, but they are probably more skewed. Below is quantile-quantile plot of the distribution of our indicators against the ones of the normal distribution.



Indeed, the distribution of our indicators is more fat-tailed compared to the one of the normal distribution. However, from our class we know that tetrad constraints to verify that there is only one underlying factor still hold under more elliptical distributions. Note, however, that tetrad constraints cannot be identified in a static way anymore, because the one factor model is dynamic. However, we can still verify them on the autocovariance structure (expand the explanation).

Table 1: Empirical Dynamic Tetrad Constraints

Lag	Tetrad 1	Tetrad 2	Tetrad 3
Lag 0	0.0524	0.0732	0.0208
Lag 1	0.0414	0.0770	0.0356
Lag 2	-0.0116	-0.0046	0.0070
Lag 3	-0.0054	-0.0031	0.0023
Lag 4	0.0066	0.0065	-0.0001
Lag 5	0.0005	0.0006	0.0001

Lag	Tetrad 1	Tetrad 2	Tetrad 3
Lag 6	0.0036	0.0041	0.0004

Even though our data is not normal, the Kalman filter is still a consistent and unbiased estimator of the underlying factor, though with non-normal data it is less efficient. We will show this in the next section though a simulation.